The Open Semantic Protocol (OSP): Decoupling Search Intelligence from Indexing via Client-Side Vector Projection

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Abstract

As Large Language Models (LLMs) become the primary interface for information retrieval, the "index" of the web is increasingly centralized within proprietary vector databases. For low-resource languages and dialects, this centralization creates a "representational collapse," where cultural nuance is lost during embedding into English-centric latent spaces. In this paper, we propose the **Open Semantic Protocol (OSP)**, a decentralized architecture that restores informational sovereignty to content creators. OSP introduces a three-layer stack: (1) **Semantic DNS** for discovering embedding data via standard TXT records; (2) **Content-Addressable Vectors** for immutable, self-hosted storage; and (3) **Just-in-Time (JIT) Semantic Projection**, a client-side mechanism that aligns disparate embedding models using lightweight linear transformation matrices. We experimentally demonstrate that a simple linear projector can bridge incompatible vector spaces (e.g., 384-dim to 768-dim) with **99.0% retrieval accuracy**, even generalizing to unseen domains (99.4% on AG News). This confirms that a decentralized, "Bring Your Own Model" search architecture is mathematically feasible and scalable.

1 Introduction

The original vision of the Semantic Web failed due to the complexity of manual ontology tagging (RDF/OWL). Today, AI-powered search engines have achieved the goal of semantic understanding, but at the cost of extreme centralization. To be "searchable" in the modern AI era, content must be ingested, embedded, and indexed by a few dominant providers (e.g., Google, OpenAI, Bing).

This centralization poses a critical threat to **Data Sovereignty**, particularly for low-resource languages and dialects (e.g., Tunisian Arabic). When a centralized indexer embeds dialectal content using a generalized English-centric model, the specific cultural and linguistic nuance is often flattened or lost. Furthermore, website owners currently have no mechanism to "host their own meaning"; they are reliant on the black-box interpretation of external crawlers.

We propose the **Open Semantic Protocol (OSP)**, a decentralized system architecture that decouples *Storage* (the Index) from *Intelligence* (the Search Engine). By leveraging existing infrastructure (DNS and HTTP/IPFS) combined with novel client-side machine learning techniques, OSP enables a web where:

1. Websites host their own vector embeddings (Self-Indexing).

[*]Code and replication scripts available at: `https://github.com/BouajilaHamza/open-semantic-protocol`

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2. Discovery is handled via standard DNS lookups (Serverless Discovery).
3. Interoperability is achieved via **Linear Projectors**, allowing a Client using Model A to search a Host using Model B without server-side compute.

2 The Open Semantic Protocol Architecture

The OSP stack consists of three distinct layers designed to run on standard web infrastructure without requiring blockchain consensus or specialized nodes.

2.1 Layer 1: Discovery (Semantic DNS)

Traditional DNS maps domain names to IP addresses. OSP extends this by mapping domain names to "Semantic Locations" using standard `TXT` records. This allows any client to discover the semantic index of a website with a single, low-latency lookup.

A typical OSP record follows the format:

```
_osp.example.com IN TXT "v=osp1; index=ipfs://QmHash...;
                 model=all-MiniLM-L6-v2"
```

This record informs the crawler or browser: (1) That the site is semantically indexed, (2) Where the index resides (IPFS CID), and (3) Which model was used to generate it.

2.2 Layer 2: Storage (Content-Addressable Vectors)

The index itself is a static file, decoupling it from compute. To ensure censorship resistance and permanence, we utilize content-addressable storage (IPFS), though standard HTTP hosting is also supported. The index file is a quantized binary or JSON structure mapping URL references to their vector representations. By using product quantization (PQ), these indices are compressed to minimal sizes, allowing for rapid client-side caching.

2.3 Layer 3: Interoperability (The Projector Layer)

The fundamental challenge of decentralized semantic search is **Model Heterogeneity**. A Host may generate embeddings using *Model H* (e.g., 384 dimensions), while a user's local browser runs *Model C* (e.g., 768 dimensions). In a naive system, these vectors are incompatible.

We solve this using **Just-in-Time (JIT) Semantic Projection**. Drawing on research in cross-lingual embedding alignment [1, 2], we posit that vector spaces describing the same semantic concepts share an isometric structure. Therefore, a linear transformation matrix W can map the Host space to the Client space.

Let $E_H(x)$ be the embedding of concept x in the Host model, and $E_C(x)$ be the embedding in the Client model. We solve for W such that:

$$E_C(x) \approx E_H(x) \cdot W \tag{1}$$

The Host publishes this matrix W (trained on a shared public corpus like Wikipedia). The Client downloads W (approx. 1-1.2MB) and applies it to the downloaded index in real-time.

3 Feasibility Study

To validate the "Projector Layer," we conducted a robust experiment aligning two architecturally distinct models: `all-MiniLM-L6-v2` (Host, 384 dimensions) and `all-mpnet-base-v2` (Client, 768 dimensions).

3.1 Experimental Setup

We trained a Ridge Regression projector (W) using 5,000 sentence pairs from the STS Benchmark (80/20 train/test split). To test robustness, we evaluated performance on two distinct datasets:

- **In-Domain:** The held-out test set of the STS Benchmark.
- **Out-of-Domain (Generalization):** The AG News dataset, which the projector never saw during training.

3.2 Results

We measured the ability of the *Client Model* to retrieve the correct document from the *Host Index* using a text query (Recall@1).

Table 1: Cross-Model Retrieval Accuracy (Recall@1)

Method	Configuration	STS Test (In-Domain)	AG News (Out-of-Domain)
No Projection	Mismatch ($384 \neq 768$)	0.0%	0.0%
OSP Projector	**Linear Alignment**	**99.0%**	**99.4%**
Oracle	Native (768)	100.0%	100.0%

The results demonstrate that a simple linear projection recovers **99%** of the native retrieval performance. Crucially, the high accuracy on AG News (99.4%) confirms that the projector learns a generalizable geometric rotation, not just a dataset-specific mapping. This implies that a single 1.15MB adapter matrix can serve a wide range of content types without retraining.

4 Related Work

llms.txt and Robots.txt: Recent proposals like `llms.txt` [3] attempt to standardize how AI agents read websites. However, these are limited to raw text and rely on centralized crawlers to perform the embedding and indexing. OSP moves the embedding step to the content owner.

Cross-Modal Alignment: Our work applies the "Connector" pattern found in Vision-Language Models (e.g., LLaVA, CLIP) [4] to the domain of network protocols. Recent work in "Mind the Gap" [5] similarly demonstrates the efficacy of linear layers in bridging code and text embeddings.

5 Discussion: Toward Sovereign AI

For nations and communities with low-resource languages, "Sovereign AI" requires control over the full stack. If the indexing layer remains centralized, the semantic representation of a culture remains subject to the biases of the indexer's model. OSP empowers Tunisian and other Arab-dialect communities to publish their own "Semantic Truth"—vectors that encode their specific cultural context—which can then be projected into global search models without loss of ownership.

6 Conclusion

The Open Semantic Protocol offers a pathway to a decentralized web where search is a protocol, not a product. By combining Semantic DNS, IPFS, and Client-Side Projection, we enable a robust, privacy-preserving, and sovereign architecture for the next generation of the AI-powered web.

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